

Real-Time 3D Graphics for Health Impact: Interactive 3D IUD Insertion

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ABSTRACT

BioDigital harnesses web-based real-time graphics to revolutionize health education, facilitating over 100 million health conversations yearly. Our content is built to be accessible to anyone with an internet connection, making health literacy available to individuals regardless of their background or location. Our Lead 3D Graphics Engineer and Content Directors will share how we built an optimized 3D interactive and enhanced engine features to expand access to reproductive healthcare in regions with unknown internet speeds and hardware specifications.

We will present a live demo of our interactive 3D visualization of an IUD Insertion procedure and detailed anatomy interactives, sharing custom rendering and asset-creation techniques developed to meet the strict technical constraints of a global audience.

CCS CONCEPTS

• **Information systems** → Web applications; • **Computing methodologies** → Scientific visualization; Interactive simulation; • **Applied computing** → Health care information systems.

KEYWORDS

Real-time rendering, Interactivity, Optimization, Scientific Visualization, Healthcare, Anatomy

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1 INTRODUCTION

Across the globe, over 200 million women want but do not have access to modern contraception [Gutmacher Institute 2022]. Due to their longevity and effectiveness, IUDs have become the global standard for safe, long-acting reversible contraception. Non-hormonal copper IUDs have the longest duration of any reversible contraception and are highly effective at preventing pregnancy for 10–12 years [Hubacher et al. 2014]. In regions with reduced access to medical facilities, IUDs are a sustainable contraceptive solution.

Despite their effectiveness, IUDs are not widely utilized in Africa [Jacobstein and Stanley 2013]. A variety of factors contribute to this low adoption, but provider confidence is a notable component. Though an IUD insertion is not more complicated than other out-patient medical procedures, a provider's confidence in performing the procedure is contingent on time, training, and inventory. All of these can be limited [bri 2014]. Minimal access to training provides fewer opportunities for providers to build confidence and skill with an IUD insertion procedure.

2 METHODS

Practice with interactive surgical content has proven to improve learners' performance with procedure comprehension [Plana et al. 2019], and studying with interactive content is more helpful than other traditional resources like textbooks, lectures, or seminars [Chaya et al. 2023]. To address the gaps in provider confidence we partnered with a global nonprofit to deliver clear and simple web-based interactive 3D content to train healthcare providers on safe IUD insertion practices.

Our cloud-based interactive platform crosses socioeconomic barriers to deliver content to any device, anywhere in the world, using only Web standards. This introduces a unique set of challenges and uncertainties, distinct from those encountered in other industries utilizing 3D interactive technology. Internet speed and hardware specifications play crucial roles in the performance of our content, and this varies significantly worldwide. In addition to performing well in the United States and Switzerland, where average download speeds exceed 200 Mbps, our content must perform equally well in places like Angola and Pakistan, where average speeds are less than 20 Mbps [int 2024].

To ensure performance across varied internet speed and hardware, the IUD Insertion interactive leverages production techniques—such as reducing mesh density, draw calls, and limiting texture sizes and keyframes [Soft8Soft 2024]. Our robust interactive anatomy models are also utilized for supplemental learning within the same constraints. With over 6,000+ individual anatomy objects in a single anatomy model, this volume poses a technical challenge for interactive performance. We employ advanced techniques within our proprietary 3D interactive engine, extending beyond traditional production methods to deliver quality interactive experiences for learners across the globe.

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